

Maclaurin Series & Convergence of Series



Constructing a Maclaurin series

- 1 Find the first four non-zero terms of the Maclaurin series for $f(x) = e^x$, using $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$.
 - 2 Find the Maclaurin series for $f(x) = \sin x$ up to the term in x^5 .
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Using known series to build new ones

- 3 Using the Maclaurin series for e^x , find the Maclaurin series for e^{-2x} up to the term in x^3 .
 - 4 Given the Maclaurin series $\sin x \approx x - \frac{x^3}{6}$ and $\cos x \approx 1 - \frac{x^2}{2}$, find the Maclaurin series for $\tan x$ up to the term in x^3 , using $\tan x = \frac{\sin x}{\cos x}$ and polynomial division.
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Using a Maclaurin series to approximate a value

- 5 Use the Maclaurin series for e^x up to the x^3 term to approximate $e^{0.1}$, to four decimal places, and compare with the true value $e^{0.1} \approx 1.10517$.
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Tests for convergence of a series

- 6 Use the ratio test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.
 - 7 Use the ratio test to determine whether the Maclaurin series for e^x , $\sum_{n=0}^{\infty} \frac{x^n}{n!}$, converges for all real x .
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Determining the radius of convergence

- 8 Use the ratio test to find the radius of convergence of the power series $\sum_{n=1}^{\infty} \frac{x^n}{n}$.
 - 9 Find the radius of convergence of $\sum_{n=0}^{\infty} n! x^n$, and explain what this tells us about where the series is useful.
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Comparing a Maclaurin approximation to the exact function

- 10 Using the Maclaurin series $\cos x \approx 1 - \frac{x^2}{2} + \frac{x^4}{24}$, approximate $\cos(0.3)$ to four decimal places, and compare with the true value $\cos(0.3) \approx 0.9553$.
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Maclaurin series for $\ln(1+x)$

- 11 Find the Maclaurin series for $f(x) = \ln(1+x)$ up to the term in x^4 , using $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4$.
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Determining the interval of convergence using the ratio test

- 12 Find the radius of convergence of the Maclaurin series for $\ln(1+x)$, $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}x^n}{n}$, using the ratio test.
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Combining Maclaurin series through multiplication

- 13 Using the Maclaurin series $e^x \approx 1 + x + \frac{x^2}{2}$ and $\frac{1}{1-x} \approx 1 + x + x^2$ (valid for small x), find the Maclaurin series for $\frac{e^x}{1-x}$ up to the term in x^2 , by multiplying the two series together.
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Using differentiation to find a Maclaurin series

- 14 Given the Maclaurin series $\frac{1}{1-x} \approx 1 + x + x^2 + x^3 + \dots$ for $|x| < 1$, differentiate both sides term by term to find the Maclaurin series for $\frac{1}{(1-x)^2}$ up to the term in x^2 .
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Estimating an error bound in a Maclaurin approximation

- 15 The Maclaurin series for $\sin x$ truncated after the x^3 term is $\sin x \approx x - \frac{x^3}{6}$. Use this to approximate $\sin(0.2)$ to four decimal places, and find the absolute error compared to the true value $\sin(0.2) \approx 0.19867$.
- 16 Find the first three non-zero terms of the Maclaurin series for $f(x) = (1+x)^{1/2}$, using the binomial series $(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2}x^2$ with $n = \frac{1}{2}$.
- 17 Use the Maclaurin series for e^x up to the x^4 term to approximate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$, and check that your answer matches the value obtained using L'Hopital's rule.

- 18 This question has three parts. (a) Find the Maclaurin series for $f(x) = \cos(x^2)$ up to the term in x^8 , by substituting x^2 into the known series $\cos u \approx 1 - \frac{u^2}{2} + \frac{u^4}{24}$. (b) Use your result to write down an approximation for $\int_0^1 \cos(x^2) dx$ as a sum of simple fractions (this integral has no elementary closed form, so series approximation is genuinely useful here). (c) Evaluate this approximation as a single decimal, to three decimal places.